



18.06 Linear Algebra

~~Linear~~ ~~Algebra~~

Lecture One)

n linear equations } where n is
 n unknowns } a variable number

ex)

$$\begin{cases} 2x - y = 0 \\ -x + 2y = 3 \end{cases}$$

$$\begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 3 \end{bmatrix}$$

A $X = B$

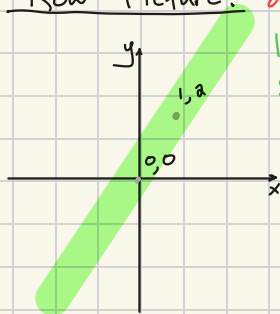
Matrix is simply a rectangular array of numbers

A - matrix of coefficients

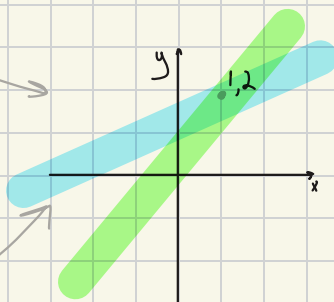
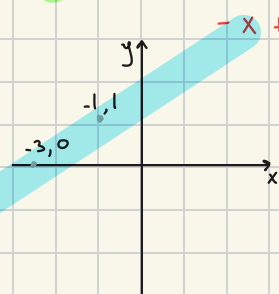
X - matrix of unknowns

B - matrix of solutions

Row Picture: $2x - y = 0$



line represents solutions to first equation



- this point lies on both solution lines

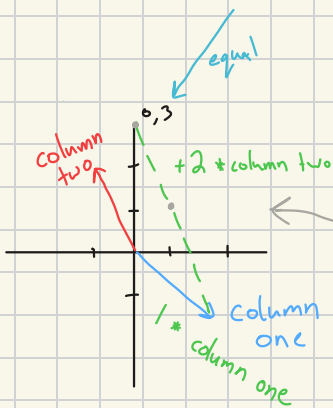
$$\begin{aligned} x &= 1 \\ y &= 2 \end{aligned}$$

Column Picture:

$$\begin{aligned} 2x - y &= 0 \\ -x + 2y &= 3 \end{aligned}$$

$$\begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 3 \end{bmatrix}$$

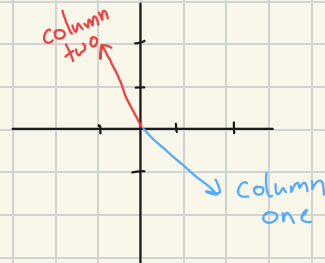
A X = B



turns
to →

linear combination

$$x \begin{bmatrix} 2 \\ -1 \end{bmatrix} + y \begin{bmatrix} -1 \\ 2 \end{bmatrix} = \begin{bmatrix} 0 \\ 3 \end{bmatrix}$$



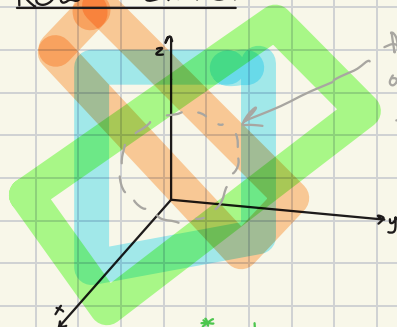
* fill in values for x, y

$$1 \begin{bmatrix} 2 \\ -1 \end{bmatrix} + 2 \begin{bmatrix} -1 \\ 2 \end{bmatrix} = \begin{bmatrix} 0 \\ 3 \end{bmatrix}$$

$$\begin{aligned} 2x - y &= 0 \\ -x + 2y - z &= -1 \\ -3y + 4z &= 4 \end{aligned}$$

$$A = \begin{bmatrix} 2 & -1 & 0 \\ -1 & 2 & -1 \\ 0 & -3 & 4 \end{bmatrix} \quad B = \begin{bmatrix} 0 \\ -1 \\ 4 \end{bmatrix}$$

Row Picture:



* each row in a 3x3 gives us a plane

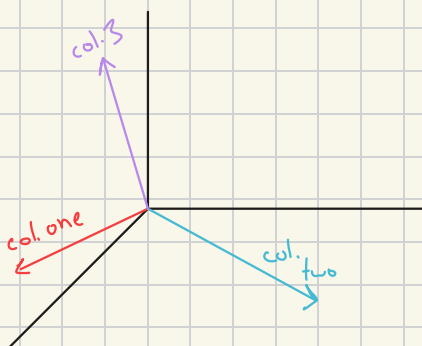
NOT CLEAR

$$x \begin{bmatrix} 2 \\ -1 \\ 0 \end{bmatrix} + y \begin{bmatrix} -1 \\ 2 \\ -3 \end{bmatrix} + z \begin{bmatrix} 6 \\ -1 \\ 4 \end{bmatrix} = \begin{bmatrix} 6 \\ -1 \\ 4 \end{bmatrix}$$

$$x=0 \quad y=0 \quad z=1$$

Column Picture:

$$x \begin{bmatrix} 2 \\ -1 \\ 0 \end{bmatrix} + y \begin{bmatrix} -1 \\ 2 \\ -3 \end{bmatrix} + z \begin{bmatrix} 6 \\ -1 \\ 4 \end{bmatrix} = \begin{bmatrix} 6 \\ -1 \\ 4 \end{bmatrix}$$



SLIGHTLY MORE CLEAR

Big Question:

can I solve $Ax = b$ for every b ?

Do the linear combinations of the columns fill 3-D space?

*how to multiply a matrix · vector

$$Ax = b$$

$$\begin{bmatrix} 2 & 5 \\ 1 & 3 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 2 \end{bmatrix} = 1 \begin{bmatrix} 2 \\ 1 \end{bmatrix} + 2 \begin{bmatrix} 5 \\ 3 \end{bmatrix} = \begin{bmatrix} 12 \\ 7 \end{bmatrix}$$

Scalar - numbers that exist by themselves with a unit of measurement.

$$x + 2y + z = 2$$

$$3x + 8y + z = 12$$

$$4y + z = 2$$

$$\begin{bmatrix} 1 & 2 & 1 \\ 3 & 8 & 1 \\ 0 & 4 & 1 \end{bmatrix} = \begin{bmatrix} 2 \\ 12 \\ 2 \end{bmatrix}$$

A — b
*solve for x or unknowns

$$Ax = b$$

FORWARD ELIM.

1st Pivot

$$\begin{array}{ccc|c} 1 & 2 & 1 & 2 \\ 3 & 8 & 1 & 12 \\ 0 & 4 & 1 & 2 \end{array}$$

$A \rightarrow U$
 $b \rightarrow c$

(2,1) step

$$\begin{array}{ccc|c} 1 & 2 & 1 & 2 \\ 0 & 2 & -2 & 6 \\ 0 & 4 & 1 & 2 \end{array}$$

x is now gone b/c 0's

(3,2) step

$$\begin{array}{ccc|c} 1 & 2 & 1 & 2 \\ 0 & 2 & -2 & 6 \\ 0 & 0 & 5 & -10 \end{array}$$

*pivots can't be zero

if zero in pivot, try to exchange for lower equation

$$\begin{array}{ccc|c} 1 & 2 & 1 & 2 \\ 0 & 2 & -2 & 6 \\ 0 & 0 & 5 & -10 \end{array}$$

U c

ELIMINATION MATRICES

Step One) Subtract 3 * row 1 from row 2

What makes ?

$$\begin{bmatrix} 1 & 2 & 1 \\ 3 & 8 & 1 \\ 0 & 4 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 1 \\ 0 & 2 & -2 \\ 0 & 4 & 1 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 \\ -3 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 1 \\ 3 & 8 & 1 \\ 0 & 4 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 1 \\ 0 & 2 & -2 \\ 0 & 4 & 1 \end{bmatrix}$$

Step Two) Subtract 2 * row 2 from row 3

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -2 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 1 \\ 0 & 2 & -2 \\ 0 & 4 & 1 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 2 & 1 \\ 0 & 2 & -2 \\ 0 & 0 & 5 \end{bmatrix}$$

BACK SUBSTIT.

new equation

$$\begin{aligned} x + 2y + z &= 2 \\ 2y - 2z &= 6 \\ 5z &= -10 \end{aligned}$$

$$\frac{5z}{5} = \frac{-10}{5}$$

$$z = -2$$

$$2y - 2(-2) = 6 \quad \text{* plug in } z$$

$$2y + 4 = 6$$

$$\frac{2y}{2} = \frac{2}{2} \quad y = 1$$

$$\begin{aligned} z &= -2 \\ y &= 1 \\ x &= 2 \end{aligned}$$

$$x + 2(1) + (-2) = 2 \quad \text{* plug in } y, z$$

$$x + 2 - 2 = 2$$

$$x = 2$$

Solving the equations in reverse order since the system is triangular.

$$E_{32} (E_{21} * A) = U$$

$$(E_{32} E_{21}) * A = U$$

* Matrix * column | column operation

$$\begin{bmatrix} 1 & 2 & 3 \\ - & - & - \\ - & - & - \\ - & - & - \end{bmatrix} \cdot \begin{bmatrix} 3 \\ 4 \\ 5 \end{bmatrix} = \begin{aligned} &3 * \text{column 1} \\ &+ \\ &4 * \text{column 2} \\ &+ \\ &5 * \text{column 3} \end{aligned}$$

* row operation | row * matrix

$$\begin{bmatrix} 1 & 2 & 7 \end{bmatrix} * \begin{bmatrix} - & - & - \\ - & - & - \\ - & - & - \end{bmatrix} = \begin{aligned} &1 * \text{row 1} \\ &+ \\ &2 * \text{row 2} \\ &+ \\ &7 * \text{row 3} \end{aligned}$$

* multiply two matrices

$$\begin{bmatrix} \text{row 3} & \text{col 4} \\ A_{31} & A_{32} \\ \vdots & \vdots \end{bmatrix} * \begin{bmatrix} B_{24} \\ B_{44} \\ \vdots \end{bmatrix} = \begin{bmatrix} C_{34} \\ \vdots \end{bmatrix}$$

$A \quad m \times n \qquad B \quad n \times p \qquad C \quad m \times p$

$$C_{34} = (\text{row 3 of } A) \cdot (\text{column 4 of } B)$$

$$= A_{31} \cdot B_{14} + A_{32} \cdot B_{24} + \dots$$

$$= \sum_{k=1}^n a_{3k} b_{k4}$$

* rows of C are combinations of the rows of B .

Column of A * a row of B

$$A \begin{bmatrix} 2 \\ 3 \\ 4 \end{bmatrix} * B \begin{bmatrix} 1 & 6 \end{bmatrix} = \begin{bmatrix} 2 & 12 \\ 3 & 18 \\ 4 & 24 \end{bmatrix} C$$

$m \times 1 \qquad 1 \times p$

* $AB = \text{sum of columns of } A * \text{rows of } B$

$$\begin{bmatrix} 2 & 7 \\ 3 & 8 \\ 4 & 9 \end{bmatrix} \begin{bmatrix} 1 & 6 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 2 \\ 3 \\ 4 \end{bmatrix} \begin{bmatrix} 16 \end{bmatrix} + \begin{bmatrix} 7 \\ 8 \\ 9 \end{bmatrix} \begin{bmatrix} 0 & 0 \end{bmatrix}$$

Block Multiplication

$$\begin{bmatrix} A_1 & A_2 \\ A_3 & A_4 \end{bmatrix} \begin{bmatrix} B_1 & B_2 \\ B_3 & B_4 \end{bmatrix} = \begin{bmatrix} A_1 B_1 + A_2 B_3 & A_1 B_2 + A_2 B_4 \\ A_3 B_1 + A_4 B_3 & A_3 B_2 + A_4 B_4 \end{bmatrix}$$

* generally all the same just different ways of illustrate.

INVERSES (square matrices) * not all matrices have inverses

$$A^{-1} * A = I$$

sqr matrix identity

if this exists...

$$A = \begin{bmatrix} 1 & 3 \\ 2 & 6 \end{bmatrix}$$

I can find a vector, x , with $A * x = 0$

$$A * x = \begin{bmatrix} 1 & 3 \\ 2 & 6 \end{bmatrix} \cdot \begin{bmatrix} 3 \\ -1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$x \neq 0$

Invertable, Non-singular

Singular, No inverse

$$\begin{bmatrix} 1 & 3 \\ 2 & 7 \end{bmatrix} \cdot \begin{bmatrix} a & c \\ b & d \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

A
 A^{-1}
 I

$A \cdot \text{column } j \text{ of } A^{-1} = \text{column } j \text{ of } I$

Gauss-Jordan (solve 2 equations at once)

$$\begin{bmatrix} 1 & 3 \\ 2 & 7 \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 3 \\ 2 & 7 \end{bmatrix} \begin{bmatrix} c \\ d \end{bmatrix} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$\left[\begin{array}{cc|cc} 1 & 3 & 1 & 0 \\ 2 & 7 & 0 & 1 \end{array} \right] \xrightarrow{A \quad I} \left[\begin{array}{cc|cc} 1 & 3 & 1 & 0 \\ 0 & 1 & -2 & 1 \end{array} \right]$$

A
 I

$$\left[\begin{array}{cc|cc} 1 & 0 & 7-3 & \\ 0 & 1 & -2 & 1 \end{array} \right]$$

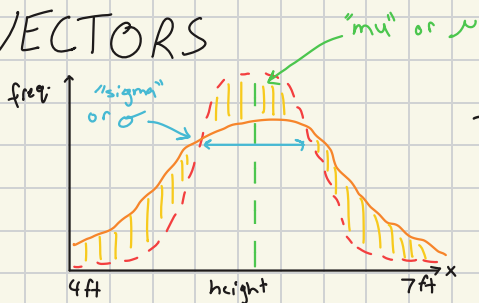
I
 A^{-1}

linear algebra is the study of vectors, vector spaces, and mapping between vector spaces.

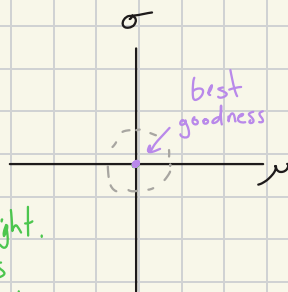
$$\begin{matrix} 2a + 3b = 8 \\ 10a + b = 13 \end{matrix} \rightarrow \begin{pmatrix} 2 & 3 \\ 10 & 1 \end{pmatrix} \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} 8 \\ 13 \end{bmatrix}$$

- problem of solving simultaneous equations and the optimization problem of fitting some data to an equation with some fitting parameters.

VECTORS



$$\rightarrow f(x) = \frac{1}{\sigma \sqrt{2\pi}} \cdot \exp \left\{ -\frac{(x-\mu)^2}{2\sigma^2} \right\}$$



* think of it like this, you get one measurement (x_1), which is the global data of height. You measure a town (x_2); if $x_1 = x_2$, it is a perfect goodness of fit (but this is rarely the case in real life). So, think of the population being represented by two parameters mean μ and standard deviation σ . For the different values of both, there will be different goodness. The closer to the origin $(0,0)$, the better goodness of fit.

a vector is an object that moves about a space; either physical or data-space. It also has two rules:

- addition
- multiplication by a scalar #

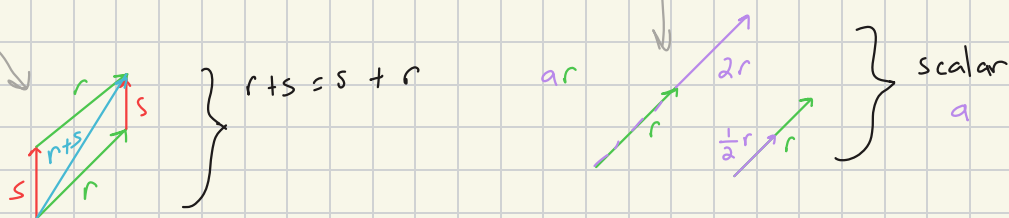
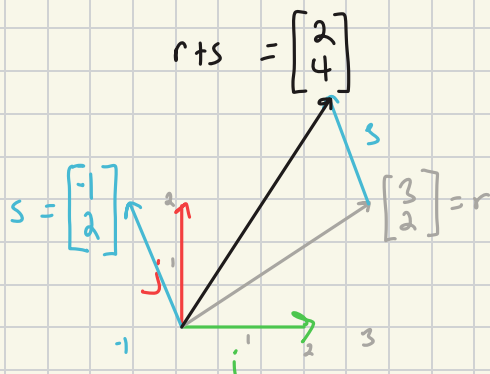


diagram)



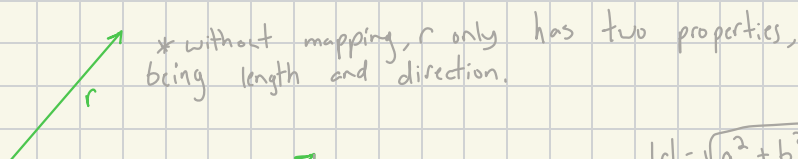
house example)

$$\text{house 1} = \begin{bmatrix} 120 \\ 2 \\ 1 \\ 150 \end{bmatrix} \begin{array}{l} \text{sqr meters} \\ \text{bedrooms} \\ \text{baths} \\ \text{\$ thousand} \end{array}$$

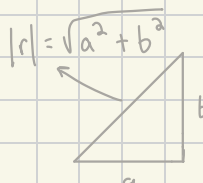
$$2(\text{house 1}) = \begin{bmatrix} 2 \cdot 120 \\ 2 \cdot 2 \\ 2 \cdot 1 \\ 2 \cdot 150 \end{bmatrix} = \begin{bmatrix} 240 \\ 4 \\ 2 \\ 300 \end{bmatrix}$$

basis vectors are what define the coordinate system used in mapping vectors (i and j in the diagram above).

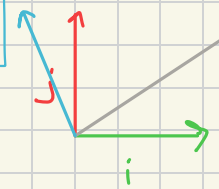
DOT PRODUCT



$$ai + bj = \begin{bmatrix} a \\ b \end{bmatrix}$$



dot product) $r \cdot s = r_i s_i + r_j s_j$

$$s = \begin{bmatrix} -1 \\ 2 \end{bmatrix} = \begin{bmatrix} s_i \\ s_j \end{bmatrix}$$


$$\begin{bmatrix} r_i \\ r_j \end{bmatrix} = \begin{bmatrix} 3 \\ 2 \end{bmatrix} = r$$

$$\rightarrow = 3(-1) + 2(2) = 1$$

the dot product has a commutative property, meaning that in the example $r \cdot s$ is equal to $s \cdot r$ (so doesn't matter direction like multiplication).

$$r = \begin{bmatrix} r_1 \\ r_2 \\ \vdots \\ r_n \end{bmatrix} \quad s = \begin{bmatrix} s_1 \\ s_2 \\ \vdots \\ s_n \end{bmatrix} \quad t = \begin{bmatrix} t_1 \\ t_2 \\ \vdots \\ t_n \end{bmatrix}$$

$$r \cdot (s+t) = r_1(s_1+t_1) + r_2(s_2+t_2) + \dots + r_n(s_n+t_n)$$

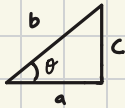
by distribution $= r_1 s_1 + r_1 t_1 + r_2 s_2 + r_2 t_2 + \dots + r_n s_n + r_n t_n$

by dot prod. $= r \cdot s + r \cdot t$

same value

* cosine rule + dot product

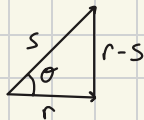
original algebra



cos rule

$$c^2 = a^2 + b^2 - 2ab \cos \theta$$

vector configuration



$$|r-s|^2 = \cancel{|r|^2} + \cancel{|s|^2} - 2|r||s|\cos \theta$$

$$(r-s) \cdot (r-s) = r \cdot r - s \cdot r - s \cdot r - s \cdot (-s)$$

$$= \cancel{|r|^2} - 2s \cdot r + \cancel{|s|^2}$$

$$+ 2s \cdot r = \cancel{+2|r||s|\cos \theta} = r \cdot s = |r||s|\cos \theta$$

!!! Full circle to cosine similarity / relations between embedding vectors

vector projection $\rightarrow r \frac{r \cdot s}{|r||r|} = r \frac{r \cdot s}{r \cdot r}$

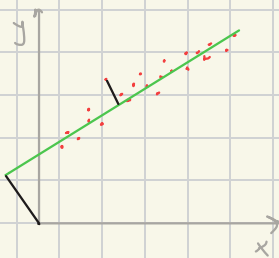
a **basis** is a set of n vectors that:

- i) are not linear combinations of each other
 - ii) span the space they describe
- , the space is then n -dimensional.

$b_3 = a_1 b_1 + a_2 b_2$ * if you can find a combination of b_1 and b_2 that equals b_3 , then it is NOT linearly independent.



example)



- good to check noise / distance from data points
- can map a data point to its dist. along the line and how far it is from the line.

* so far have covered the following:

- vectors; objects that describe where we are in space (physical, dataspace, parameters)
- defined vector addition and scaling
- found mod of vectors and the dot product
- defined a vector spaces dimension
- linear independence, dependence, combinations.

MATRICES

* multiply matrix by x-axis

$$\begin{bmatrix} 2 & 3 \\ 10 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 \\ 10 \end{bmatrix}$$

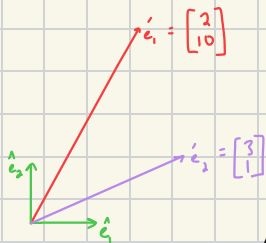
* now by y-axis

$$\begin{bmatrix} 2 & 3 \\ 10 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 1 \end{bmatrix}$$

the set of simultaneous equations $\begin{bmatrix} 2 & 3 \\ 10 & 1 \end{bmatrix}$ is asking what vector is needed to get an output of $\begin{bmatrix} 8 \\ 13 \end{bmatrix}$.

$$\begin{aligned} 2a + 3b &= 18 \\ 10a + 1b &= 13 \end{aligned}$$

$$\begin{bmatrix} 2 & 3 \\ 10 & 1 \end{bmatrix} \begin{bmatrix} a \\ b \end{bmatrix} = \begin{bmatrix} 8 \\ 13 \end{bmatrix}$$



moving up or down on $\hat{e}_1$ and $\hat{e}_2$ is changing the variables a and b . The goal is to find where they add up to $\begin{bmatrix} 8 \\ 13 \end{bmatrix}$.